Results from a test to calibrate the difference in rotational speed of chassis‘s motors

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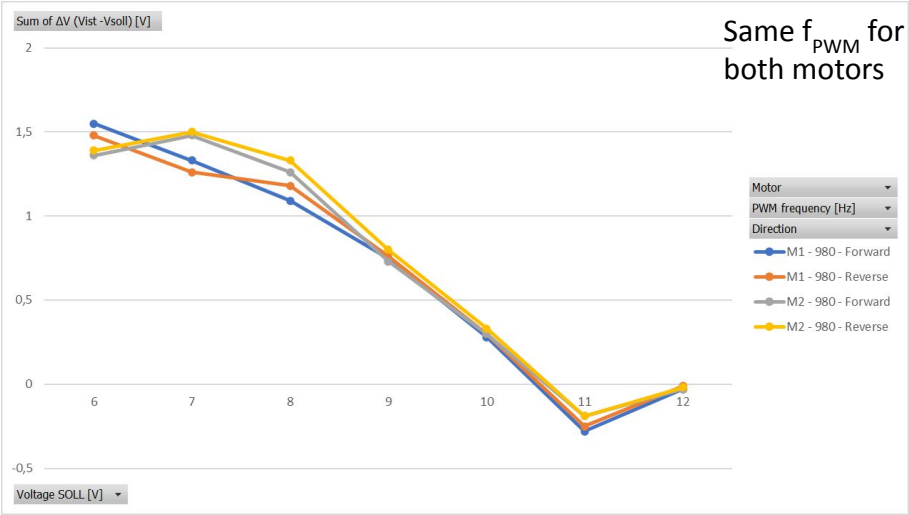
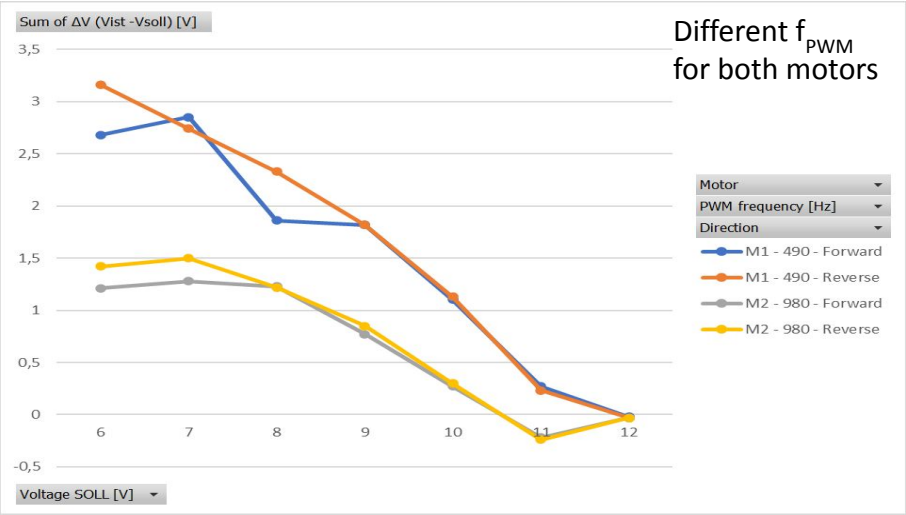
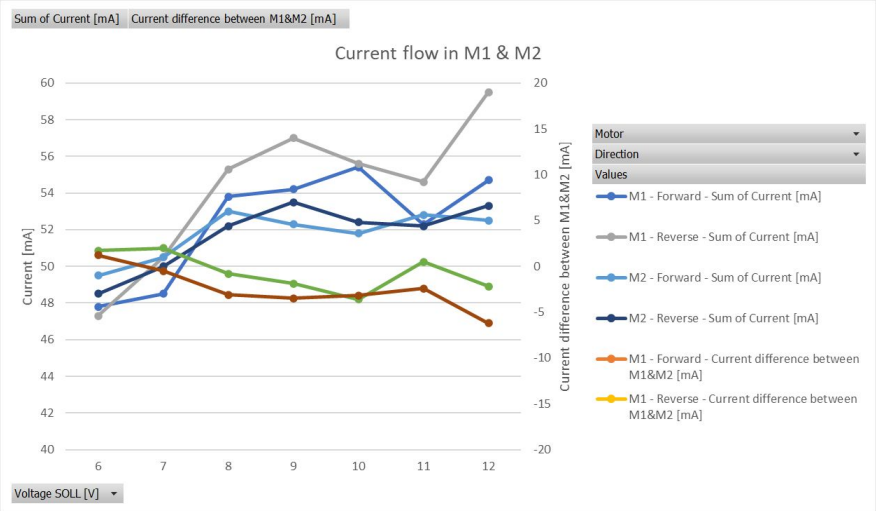
Difference between V_{IST} & V_{SOLL} in M1&M2



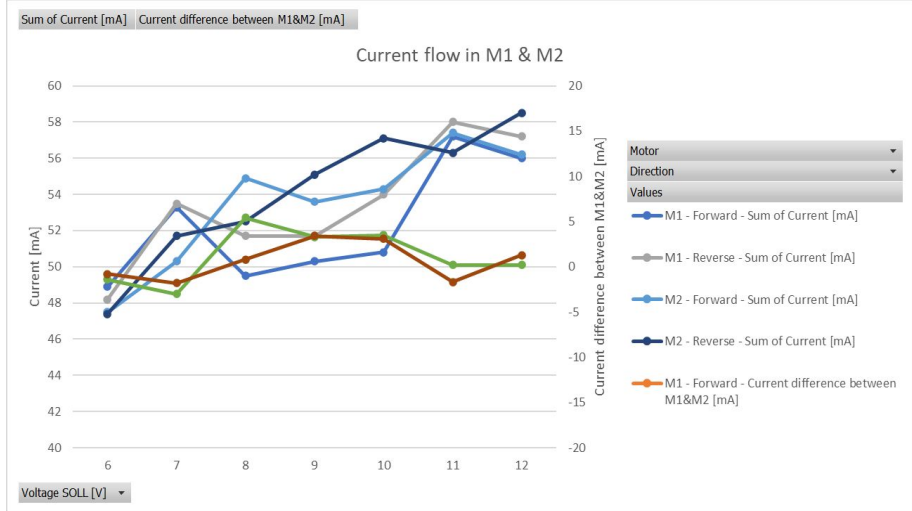


1. For a same/fixed duty cycle (D), the voltage output from motor driver is dependant on the PWM frequency (f_{PWM}).
2. Low f_{PWM} accounted for high voltage output from the motor driver. This cause a high inaccuracy of V_{IST} from V_{SOLL}.
3. Conclusion:
   a. Get the correct f_{PWM} in order to get the desired voltage ouput on motor driver.
   b. Duty cycle, D< 50% is sensitive to this phenomenon. See [Result - Calibration straight maneuver.xlsx]

Current flow in M1& M2



Same f_{PWM} for both motors. M1 = 980Hz; M2 = 980Hz



Different f_{PWM} for both motors. M1 = 490Hz; M2 = 980Hz

1. For different $f_{PWM,}$ the current difference between M1 & M2 at V_{SOLL} = 8V is higher than the same f_{PWM}
2. Conclusion:
   a. No significant indication for the influence of f_{PWM} on the current flow in M1&M2